



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Data collected via SpaceX API and Wikipedia scraping
- EDA performed using visualization and SQL
- Interactive analysis with Folium and Dash
- ML models built for prediction
- Source: API + Wrangling + EDA + ML notebooks



Introduction

SpaceX has revolutionized the aerospace industry by significantly reducing the cost of rocket launches through the reuse of Falcon 9 first-stage boosters. A Falcon 9 launch costs approximately \$62 million, compared to over \$165 million for traditional providers. The ability to successfully land and reuse boosters is the key factor behind these cost savings

- **Problem to solve:**

Predict the successful landing of SpaceX Falcon 9 first-stage boosters to estimate launch cost efficiency.

- **Why?**

- Accurate predictions enable companies to:
 - Estimate launch costs more effectively
 - Make competitive bids in the aerospace market

Section 1

Methodology

Methodology

Executive Summary

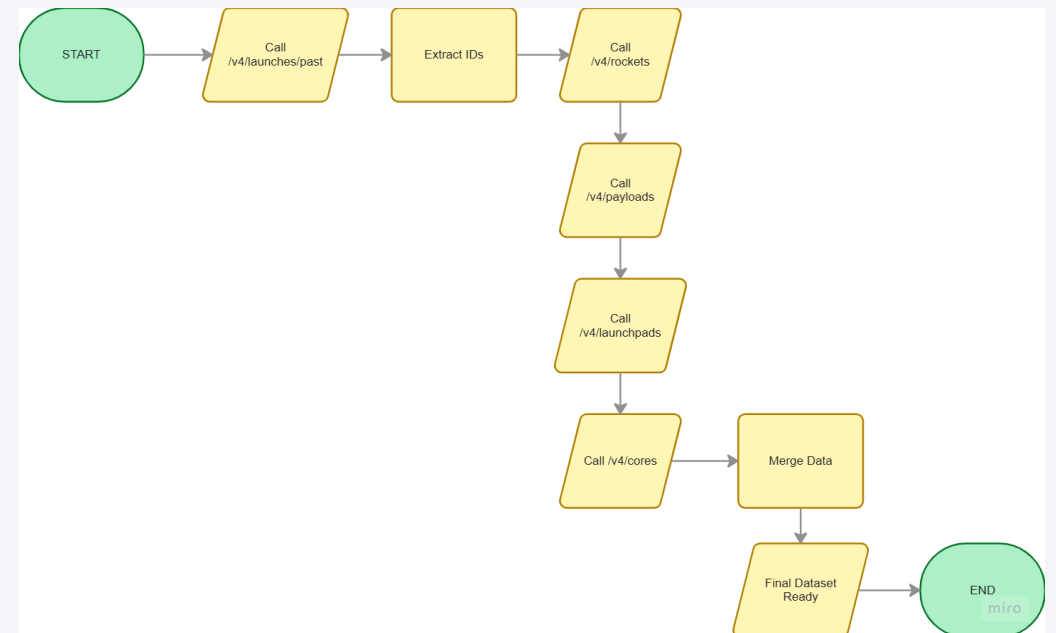
- Data collection methodology:
 - Data was collected using SpaceX REST API endpoints and web scraping from Wikipedia to build a comprehensive dataset.
- Perform data wrangling
 - Data was cleaned by handling missing values, transforming variables, and creating a binary target variable for landing success
- Perform exploratory data analysis (EDA) using visualization and SQL
 - EDA was performed using visualization and SQL to identify patterns between variables such as payload, orbit, and launch site
- Perform interactive visual analytics using Folium and Plotly Dash
 - Interactive maps and dashboards were created using Folium and Plotly Dash to explore launch locations and success outcomes dynamically.
- Perform predictive analysis using classification models
 - Classification models (Logistic Regression, SVM, Decision Tree, KNN) were built, tuned, and evaluated to predict landing success.

Data Collection

- Data was collected from multiple sources:
 - Initial dataset was obtained from /v4/launches/past
 - APIs were used to retrieve structured data
 - Web scraping was applied for additional data
 - Raw data was extracted using Python tools
 - Data was gathered in real-time or batch processes

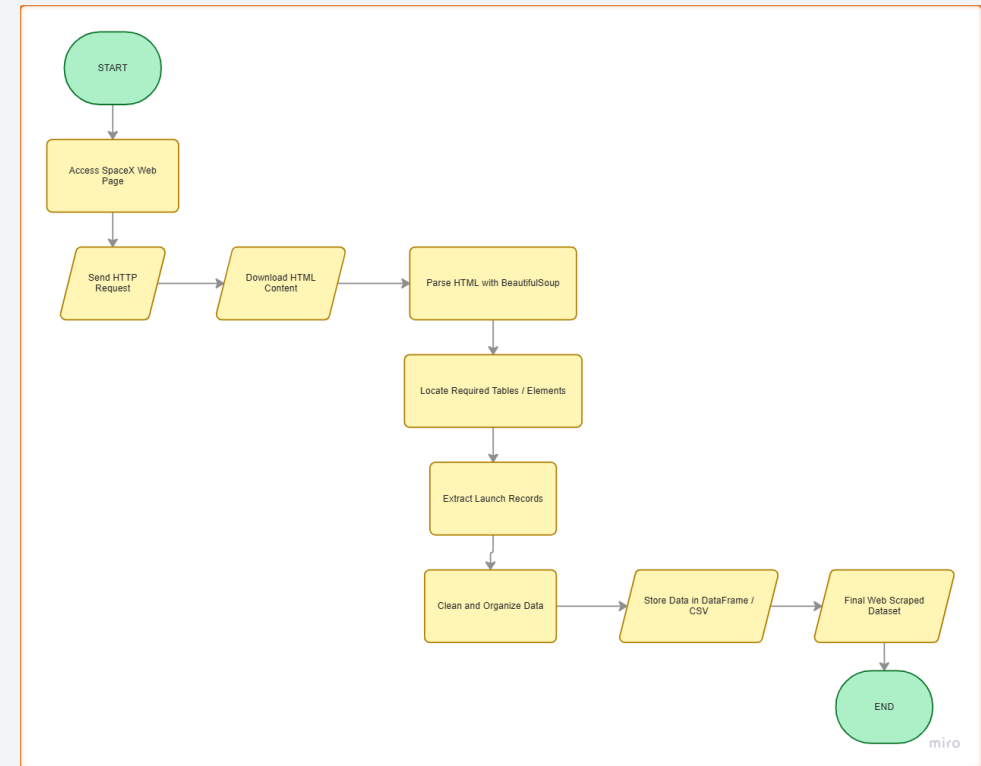
Data Collection – SpaceX API

- Initial data was obtained from the `/v4/launches/past` API endpoint.
- GitHub [Notebook](#)



Data Collection - Scraping

- **Web Scraping**
 - Web Page
 - HTTP Request
 - HTML Data
 - Parse (BeautifulSoup)
 - Extract Data
 - Clean Data
 - Store Dataset

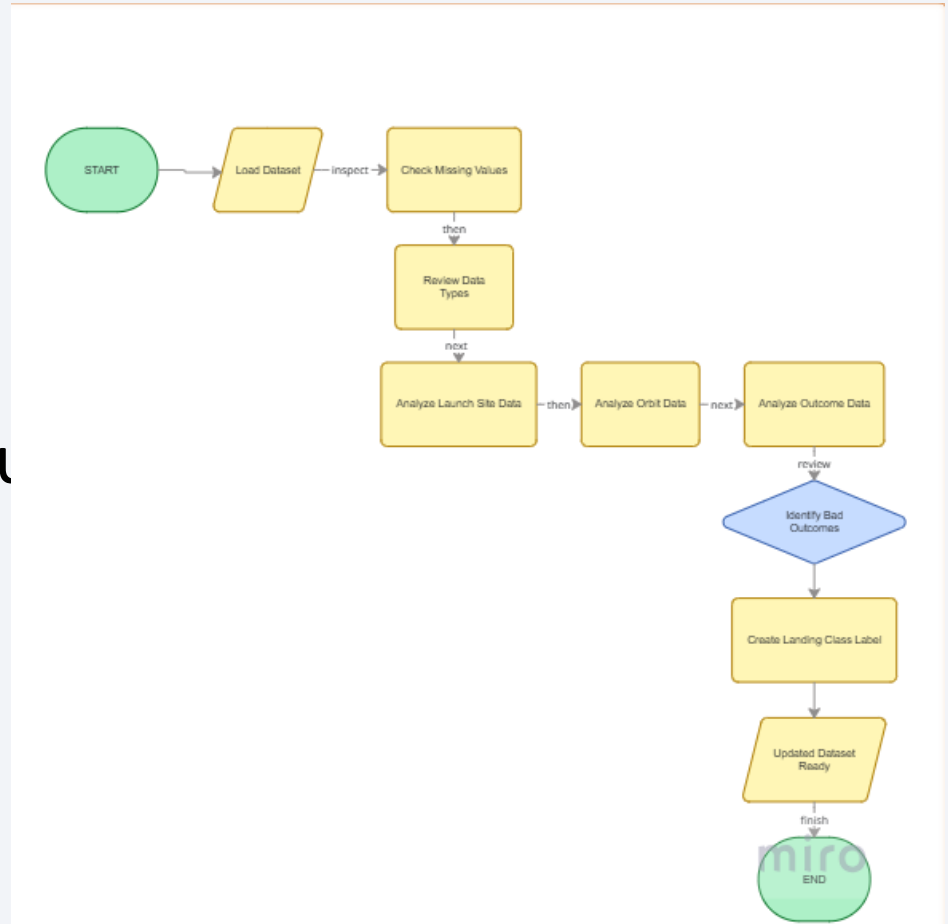


GitHub [Notebook](#)

Data Wrangling

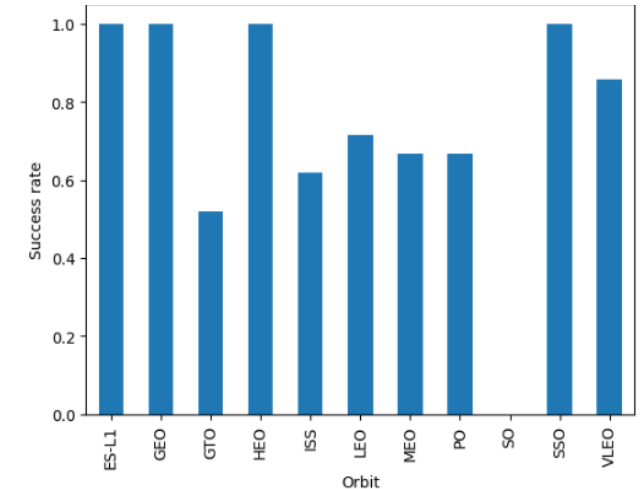
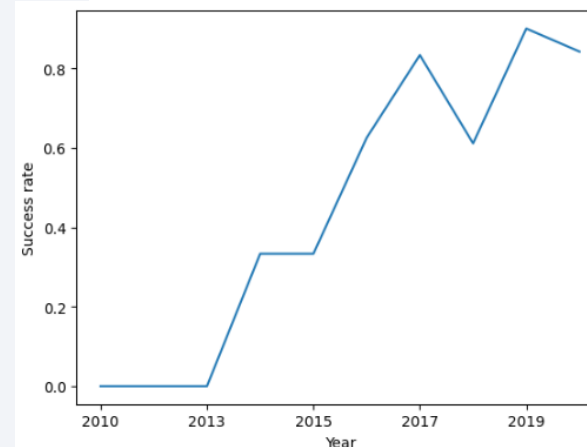
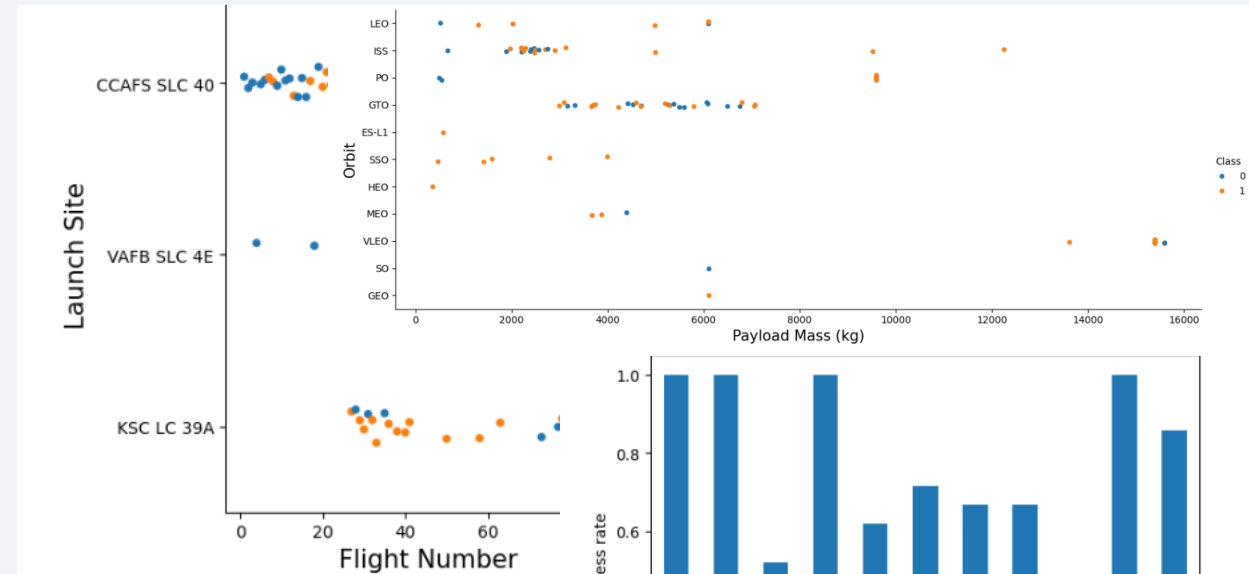
- Load the dataset
- Check missing values
- Review data types
- Analyze launch site, orbit, and outcome columns
- Identify unsuccessful landing outcomes
- Create the Class label
- Prepare the final wrangled dataset

GitHub [Notebook](#)



EDA with Data Visualization

- Flight Number vs Outcome/Site: Identifies trends in landing success over time and across launch sites
- Payload vs Launch Site: Compares payload distribution to understand differences between launch locations.
- Success Rate vs Orbit Type: Highlights which orbit types have higher landing success rates.
- Flight Number vs Orbit Type: Shows how mission profiles evolve with increasing flight numbers.
- Payload vs Orbit Type: Examines the interaction between payload mass and orbit type.
- Yearly Success Trend: Demonstrates improvement in landing success over time.

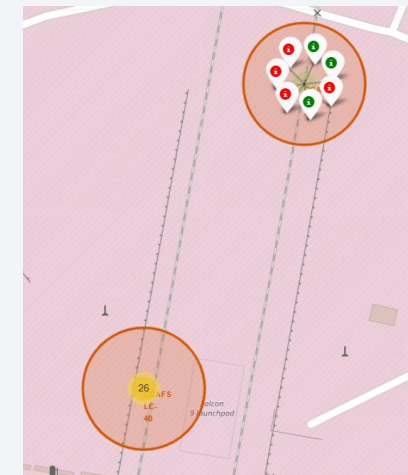
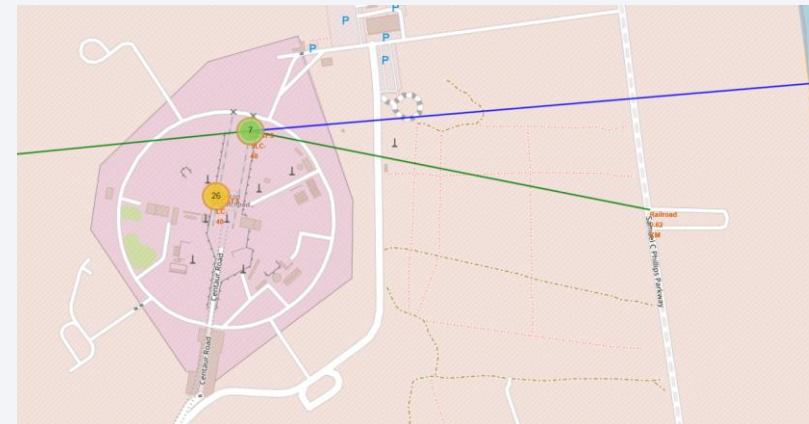
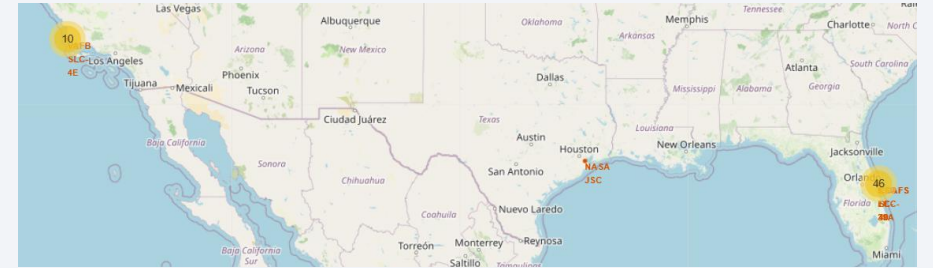


EDA with SQL

- Retrieved distinct launch sites using SELECT DISTINCT
 - Filtered records by launch site patterns using WHERE and LIKE
 - Calculated total and average payload using SUM and AVG
 - Identified first successful landing using MIN and date filtering
 - Selected boosters based on payload ranges using WHERE conditions
 - Counted success and failure outcomes using COUNT and GROUP BY
 - Retrieved boosters with maximum payload using subqueries and MAX
 - Filtered and analyzed specific missions (e.g., year 2015) Ranked landing outcomes using GROUP BY and ORDER BY
-
- **GitHub [Notebook](#)**

Build an Interactive Map with Folium

- Markers (Launch Sites): Added to display the geographical locations of all launch sites.
- Colored Markers (Success/Failure): Used to distinguish successful (green) and failed (red) landings.
- Circles: Highlight launch site areas and improve visual emphasis.
- Lines (Distance Lines): Drawn to show distances between launch sites and nearby infrastructures (coast, roads, railways).



Build a Dashboard with Plotly Dash

A dashboard was created to explore the data interactively

- A dropdown menu allows selecting a launch site:
 - Pie chart updates to show success distribution
 - Shows overall success (all sites) or success vs failure (selected site)
 - A scatter plot displays the relationship between payload mass and launch outcome
 - Updates based on selected launch site
 - A payload mass range slider filters the data shown in the scatter plot

GitHub [Notebook](#)

Predictive Analysis (Classification)

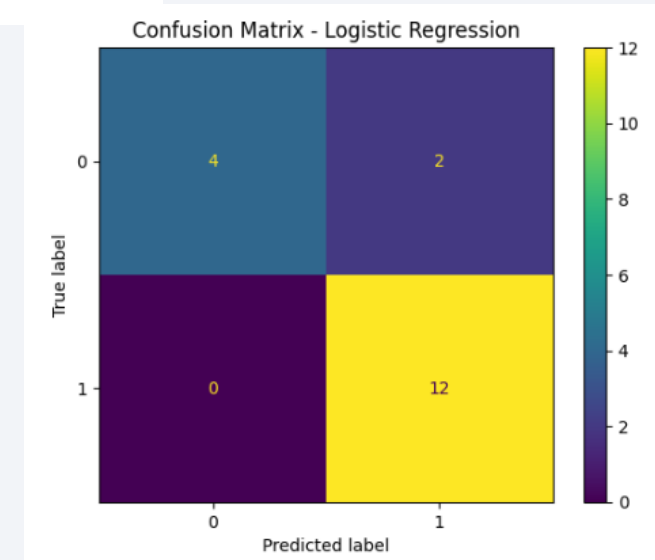
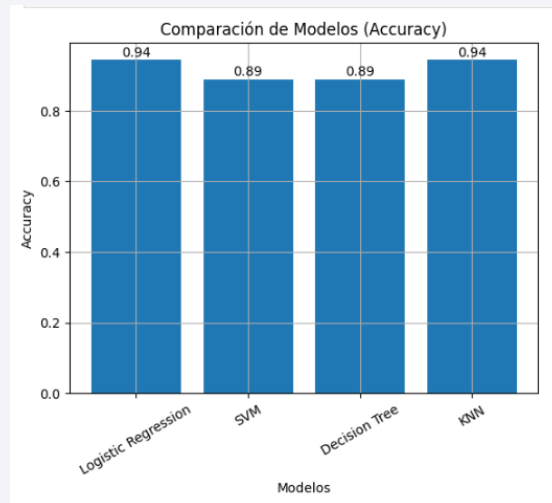
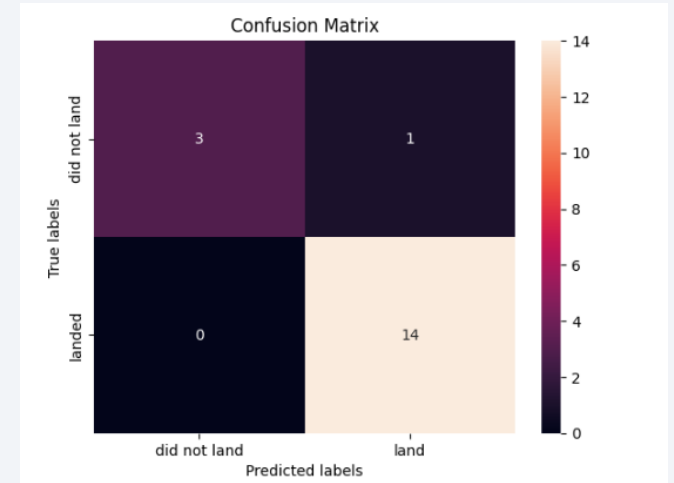
- The dataset was loaded and the target variable Class was defined
- Features were standardized and split into train and test sets
- Four models were trained using GridSearchCV:
 - Logistic Regression
 - SVM
 - Decision Treea
 - KNN
- Model performance was evaluated with accuracy and confusion matrices to identify the best classifier

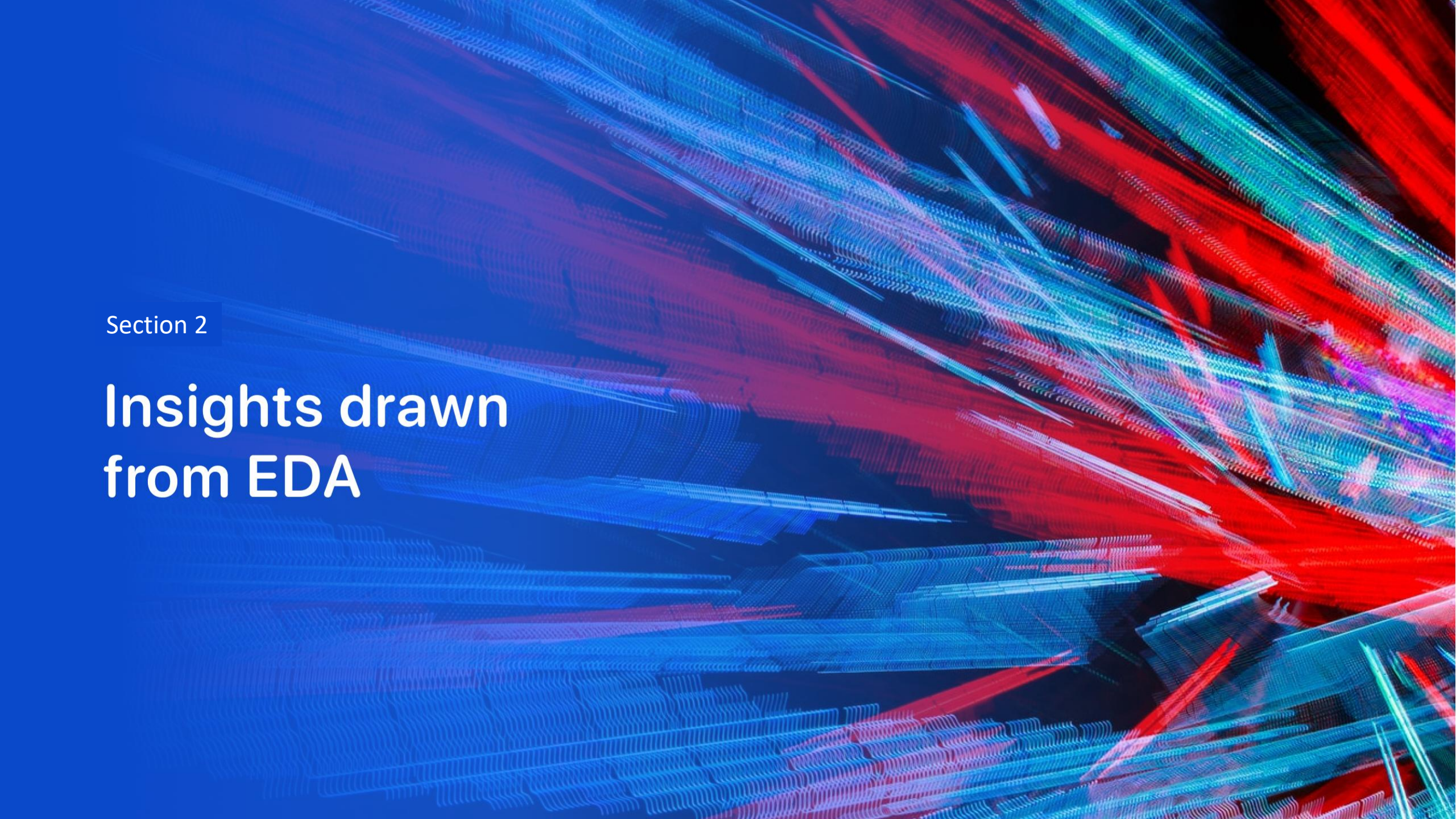
GitHub [Notebook](#)

Results

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...	Serial_B1058	Serial_B1059	Serial_B1060	Serial_B1062	C
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0	0.0	0.0	
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
85	86.0	15400.000000	2.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	
86	87.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	1.0	0.0	0.0	0.0	
87	88.0	15400.000000	6.0	5.0	5.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	
88	89.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	
89	90.0	3681.000000	1.0	5.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0	

90 rows × 83 columns



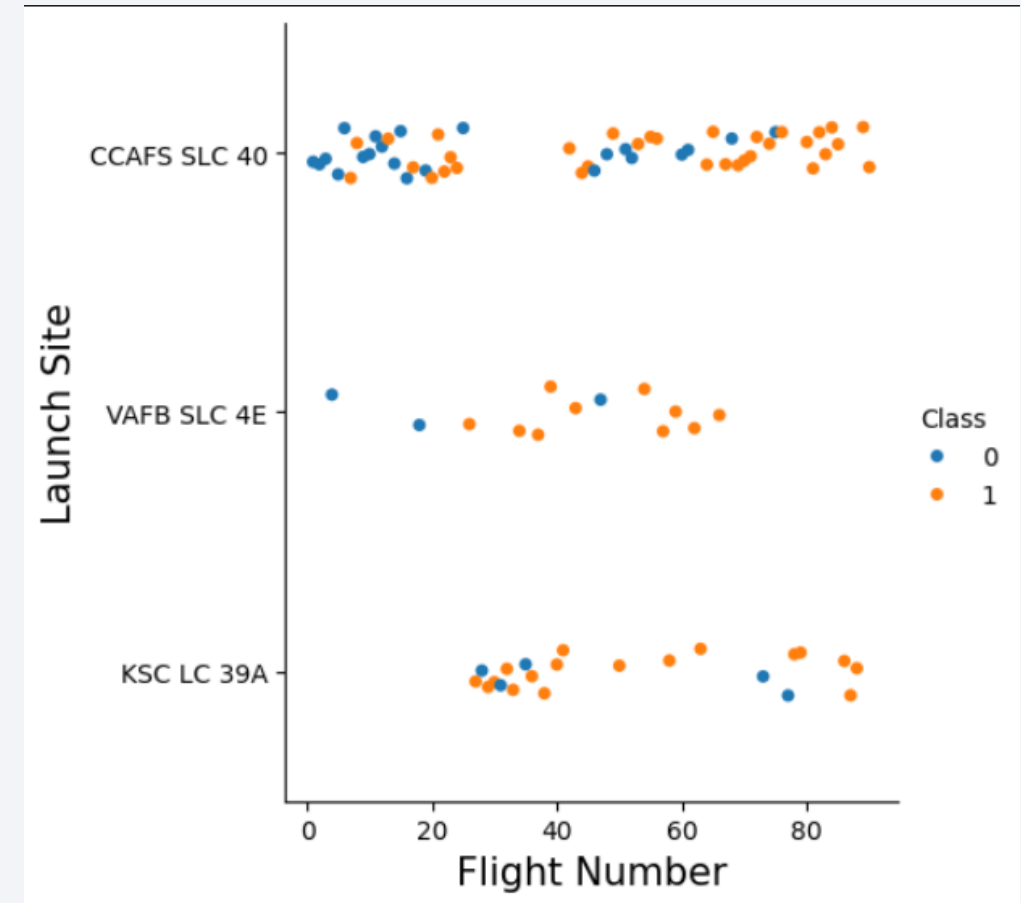
The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

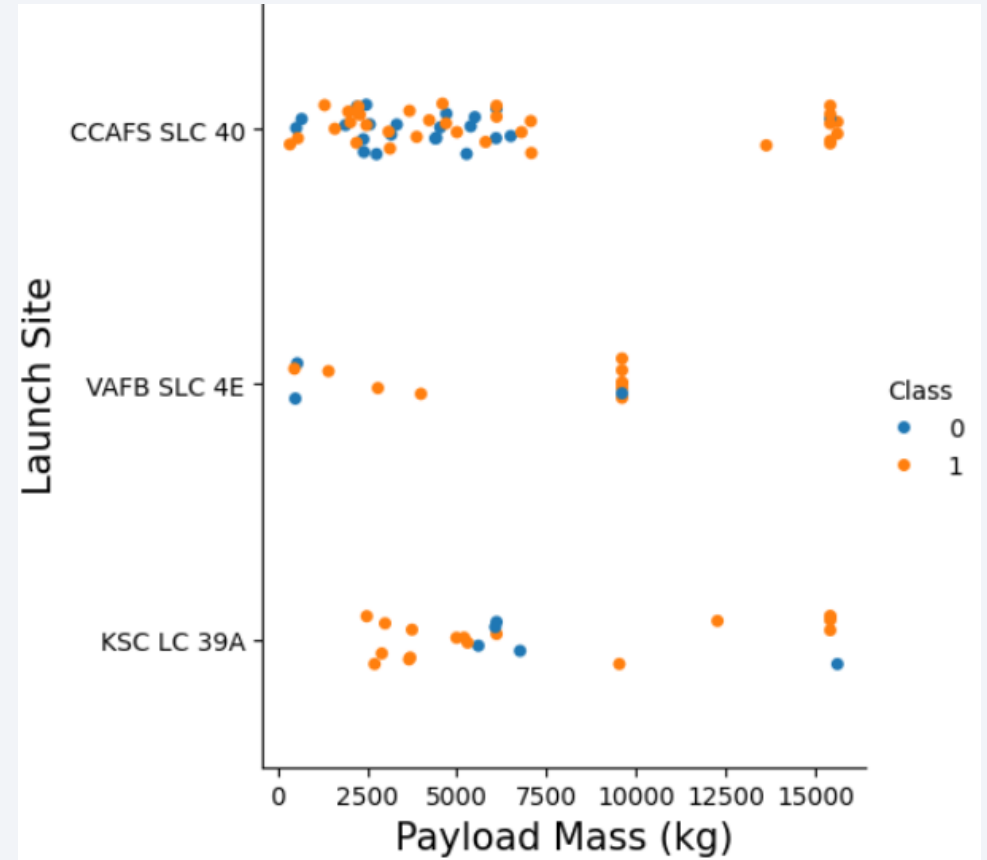
Flight Number vs. Launch Site

- Launch success (Class = 1) increases with flight number
- CCAFS SLC 40 shows higher consistency in successful landings
- KSC LC 39A also presents many successful outcomes
- VAFB SLC 4E shows more variability and fewer successes



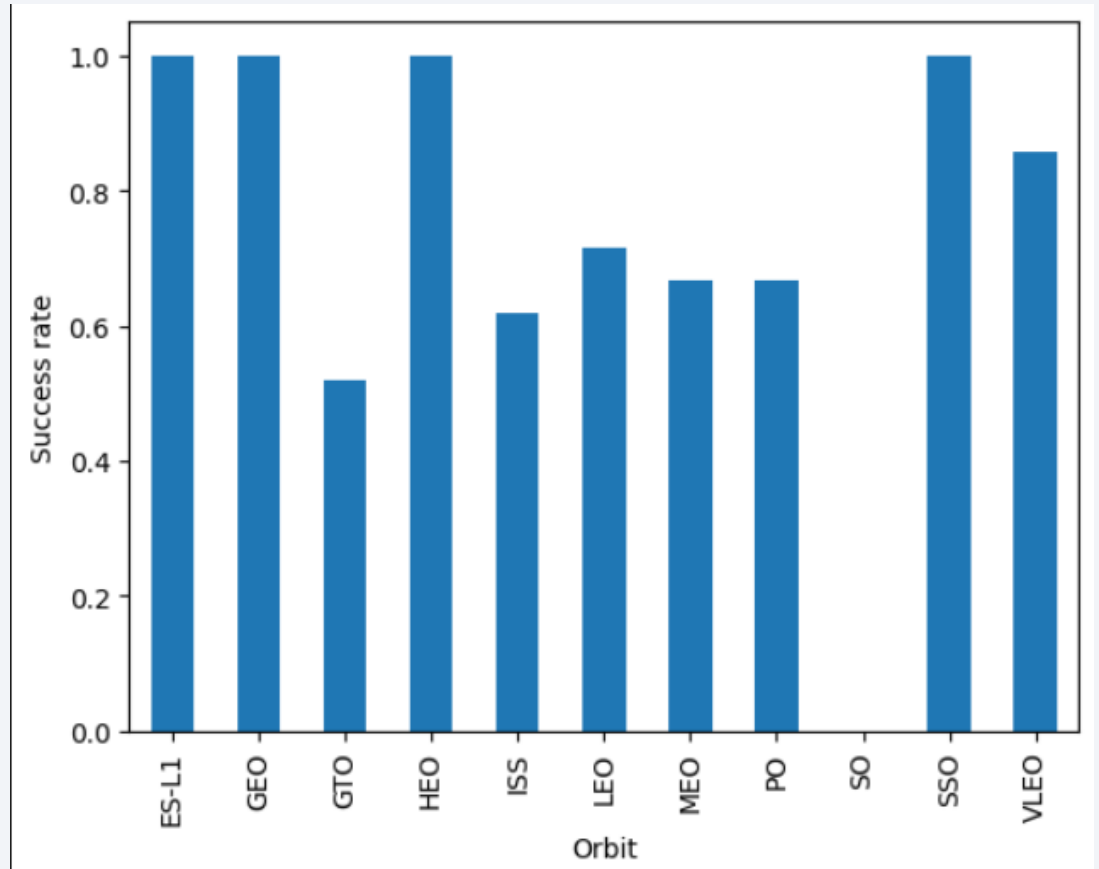
Payload vs. Launch Site

- Higher payload masses tend to be associated with successful landings (Class = 1)
- CCAFS SLC 40 shows a high concentration of successful launches across different payloads
- KSC LC 39A presents good performance, especially at medium payload ranges
- VAFB SLC 4E shows fewer launches and more variability in outcomes



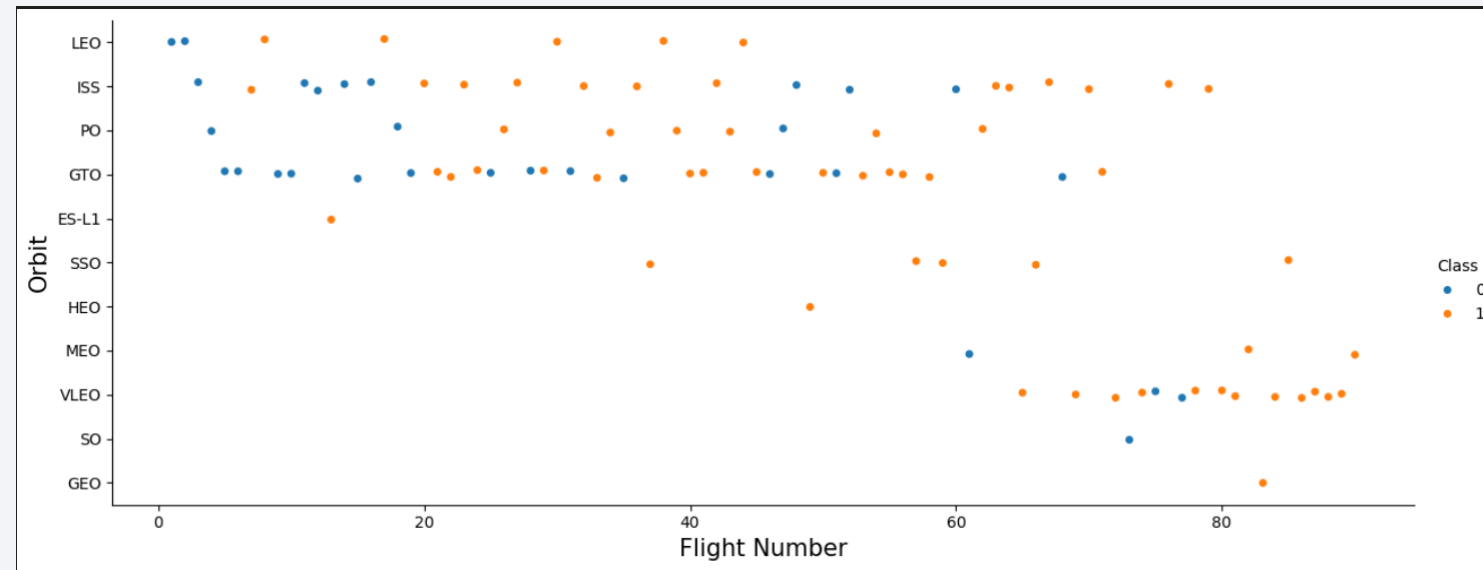
Success Rate vs. Orbit Type

- Some orbits (ES-L1, GEO, HEO, SSO) show very high success rates (close to 100%)
- GTO orbit presents a lower success rate compared to others
- LEO, MEO, and ISS orbits show moderate and stable performance
- Success rate varies significantly depending on the orbit type



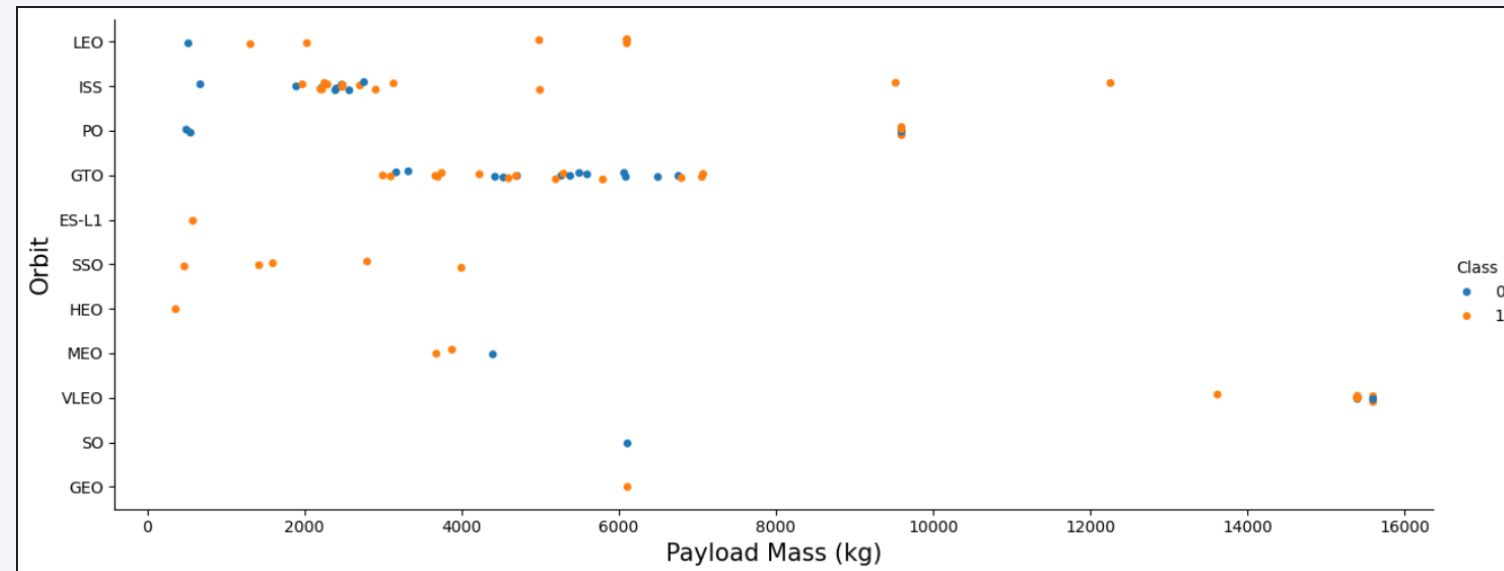
Flight Number vs. Orbit Type

- Launch success increases with flight number across most orbits
- Early flights show more failures (Class = 0), later flights more successes (Class = 1)
- Some orbits (LEO, ISS, GTO) have a higher concentration of successful outcomes
- Booster performance improves over time regardless of orbit type



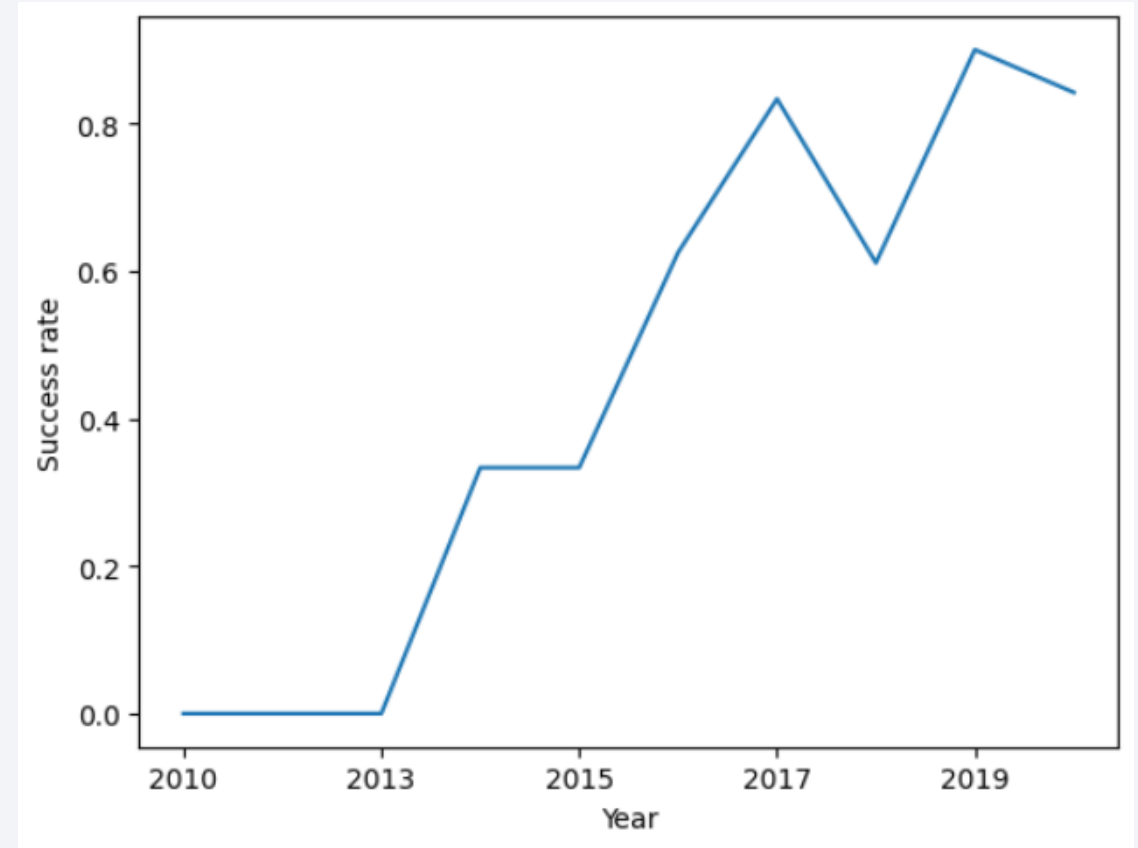
Payload vs. Orbit Type

- Successful landings (Class = 1) occur across a wide range of payload masses
- Medium payload ranges (≈ 2000 – 6000 kg) show a high concentration of launches
- High payload masses ($> 10,000$ kg) are mostly associated with successful outcomes
- Orbit type influences results, but success is achieved across multiple orbits



Launch Success Yearly Trend

- Success rate increases significantly over time
- Early years (2010–2013) show no successful landings
- Improvements begins around 2014–2016
- High success rates are achieved from 2017 onwards



All Launch Site Names

- %%sql **SELECT DISTINCT** Launch_Site **FROM** SPACEXTABLE;

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

- **SELECT *FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;**

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- **SELECT SUM(PAYLOAD_MASS__KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer LIKE 'NASA (CRS)%';**

Total_Payload_Mass
48213

Average Payload Mass by F9 v1.1

- **SELECT AVG(PAYLOAD_MASS__KG_) AS Avg_Payload_Mass
FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';**

Avg_Payload_Mass

2928.4

First Successful Ground Landing Date

- **SELECT MIN(Date) AS First_Success_Ground_Pad_Date FROM SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (ground pad)';**

First_Success_Ground_Pad_Date
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- **SELECT** Booster_Version **FROM** SPACEXTABLE
WHERE "Landing_Outcome" = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000
AND PAYLOAD_MASS__KG_ < 6000;

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- **SELECT "Mission_Outcome", COUNT(*) AS Count
FROM SPACEXTABLE
GROUP BY "Mission_Outcome";**

Mission_Outcome	Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- **SELECT** Booster_Version **FROM** SPACEXTABLE
WHERE PAYLOAD_MASS__KG_ = (
SELECT **MAX**(PAYLOAD_MASS__KG_)
FROM SPACEXTABLE
-);

Booster_Version

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

2015 Launch Records

- **SELECT** substr(Date, 6, 2) **AS Month**, "Landing_Outcome",
Booster_Version, Launch_Site
FROM SPACEXTABLE
WHERE substr(Date, 1, 4) = '2015'
AND "Landing_Outcome" = 'Failure (drone ship)';

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- **SELECT "Landing_Outcome", COUNT(*) AS Count**
FROM SPACEXTABLE
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY "Landing_Outcome"
ORDER BY Count DESC;

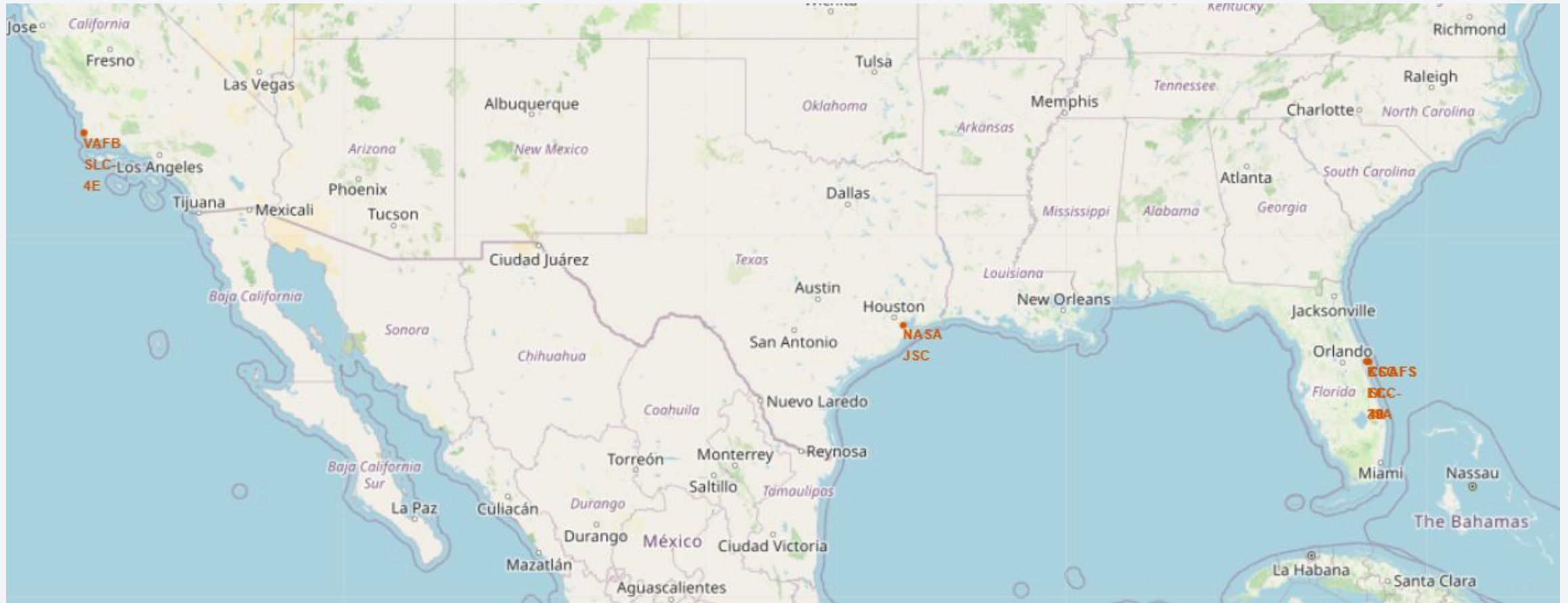
Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

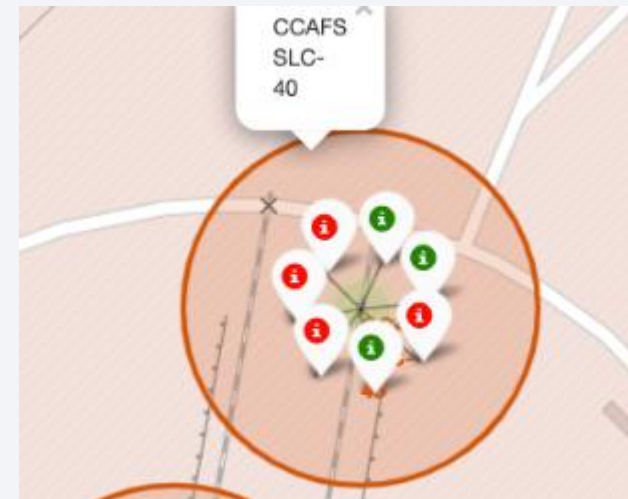
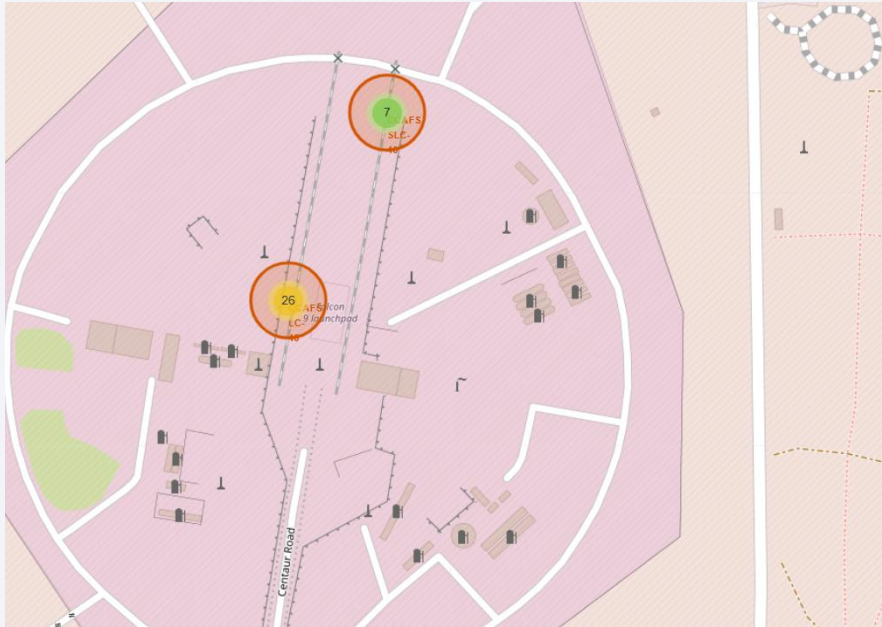
Section 3

Launch Sites Proximities Analysis

Launch site locations

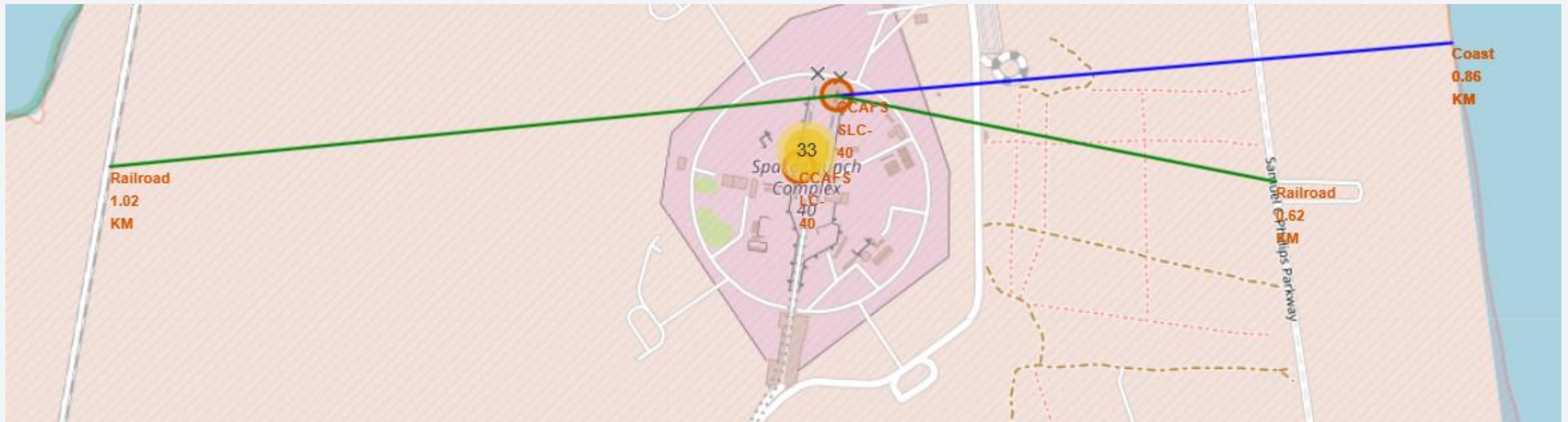


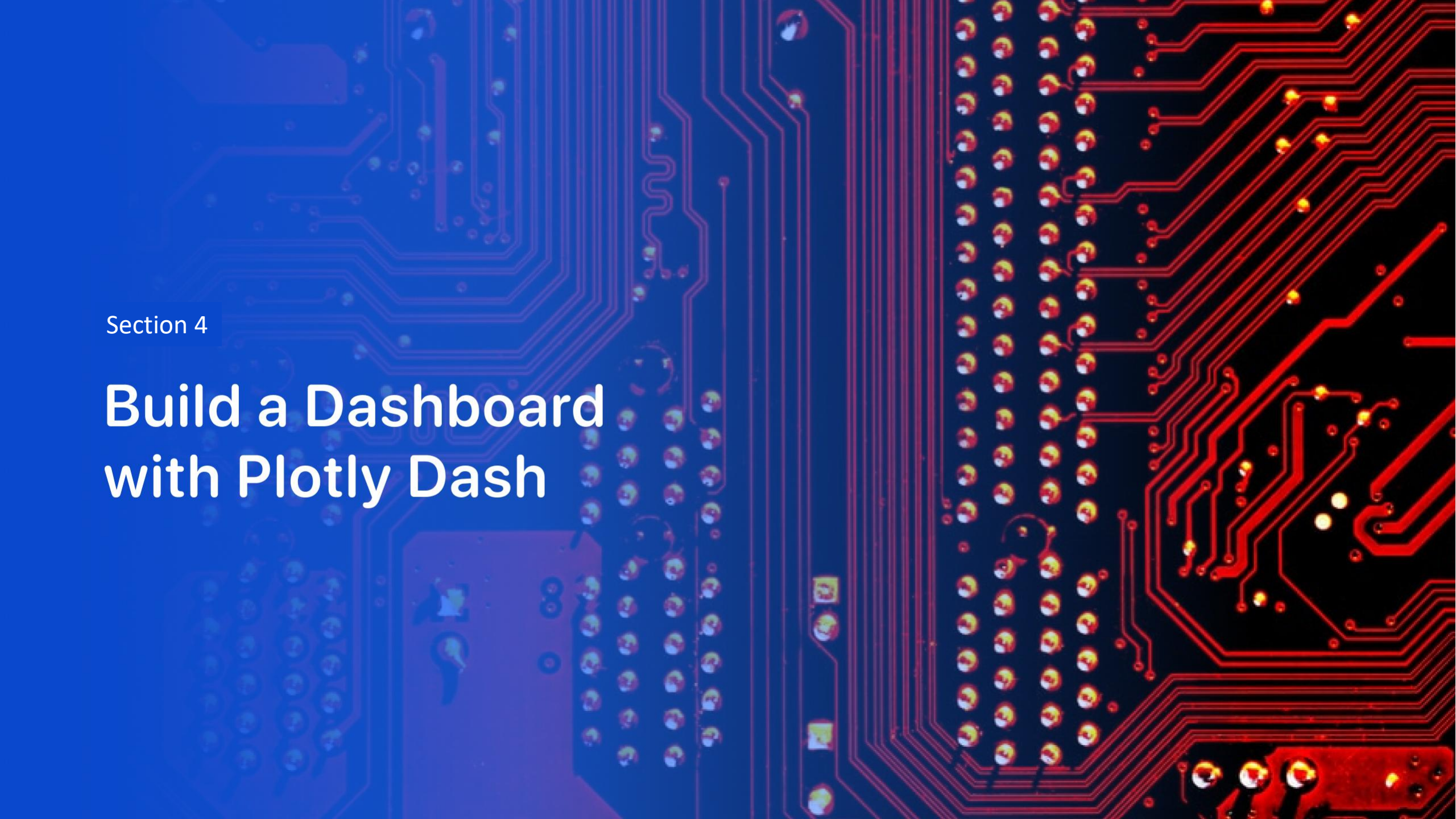
Launch Outcomes



Near Locations

- The launch site (CCAFS SLC 40) is located very close to the coast (~0.86 km)
- Distance to infrastructure such as railroads varies between ~0.6–1.0 km
- Launch sites are strategically placed near the coastline for safety reasons
- Proximity to transport infrastructure supports logistics and operations





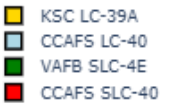
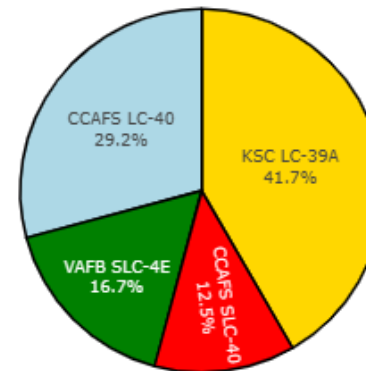
Section 4

Build a Dashboard with Plotly Dash

Successful Landings of all sites

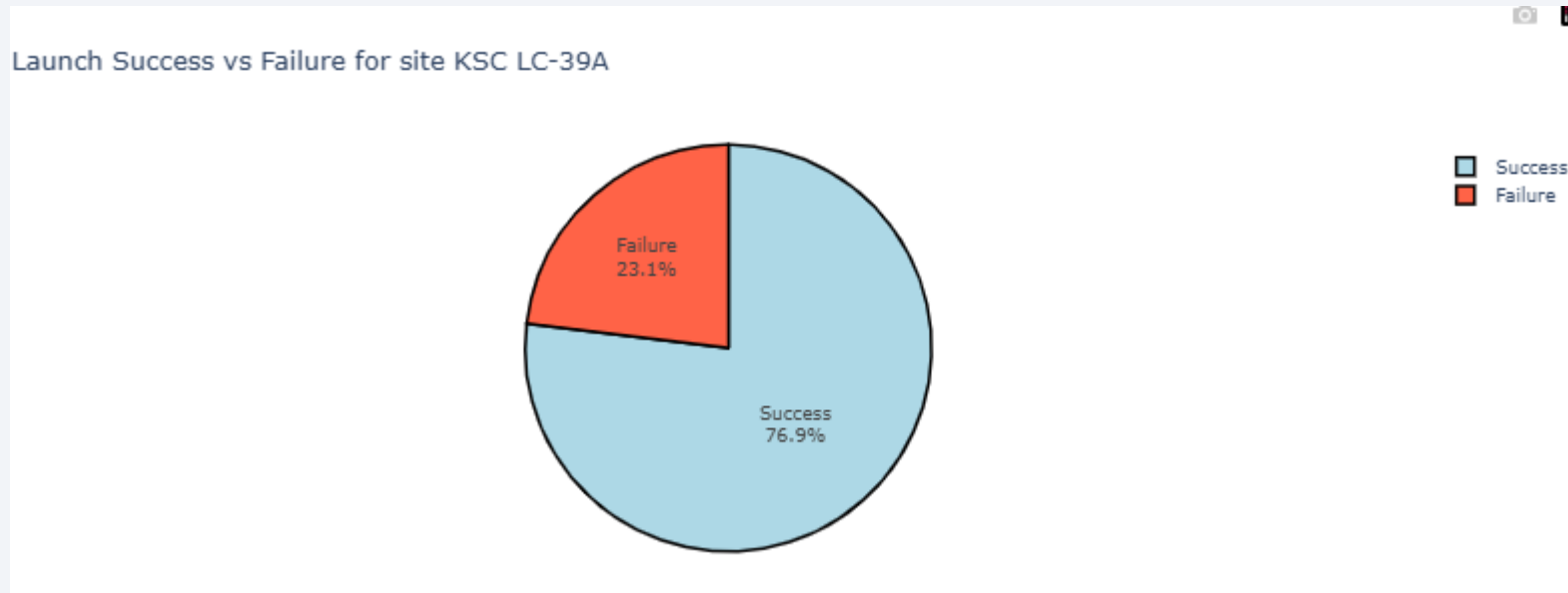
- KSC LC-39A has the highest number of successful launches (41.7%)
- CCAFS LC-40 is the second most successful site (29.2%)
- VAFB SLC-4E contributes a smaller share (16.7%)

Total Successful Launches by Site



HIGH Success Ratio

- KSC LC-39A shows a high success rate (76.9%)
- Failures represent a smaller proportion (23.1%)



Payload (2000-5000)

- Newer booster versions (B4, B5) are associated with higher success rates
- FT shows a mix of successful and failed launches across the payload range
- v1.1 presents more failed outcomes, especially at higher payload values

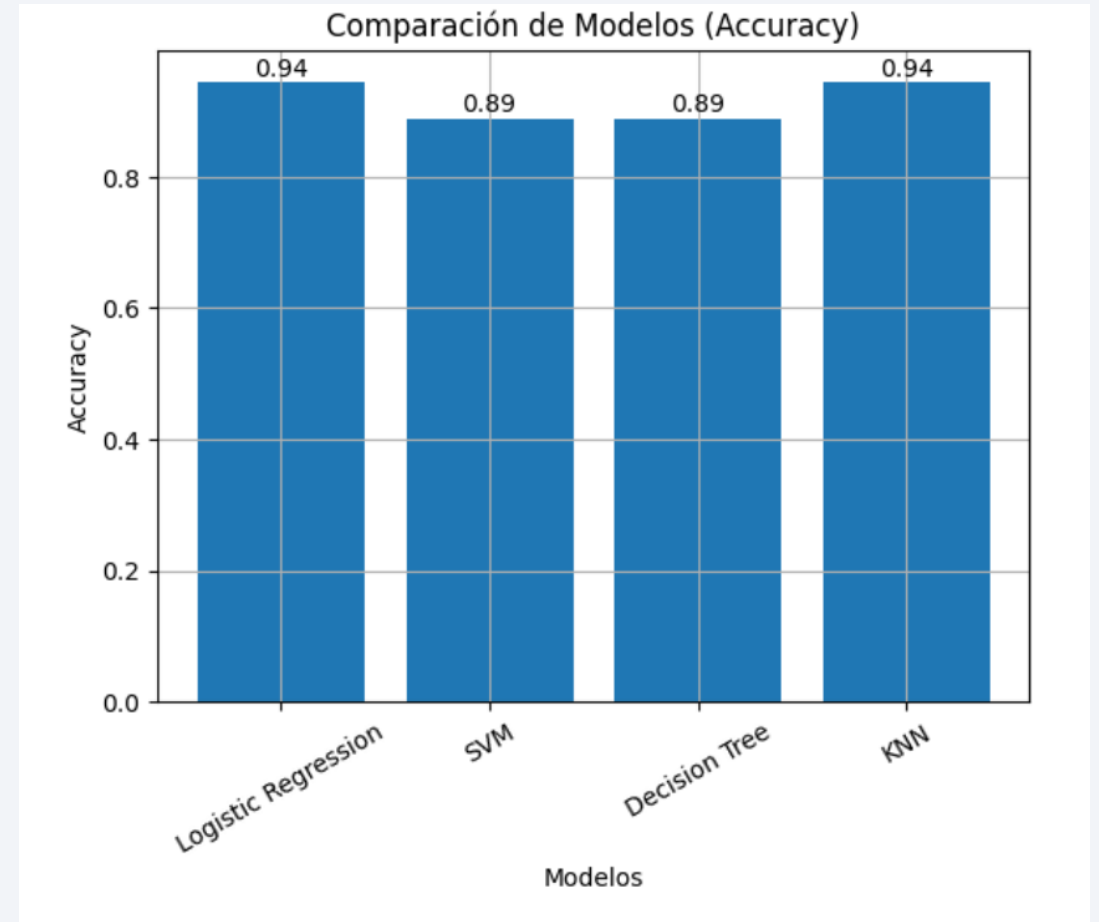


Section 5

Predictive Analysis (Classification)

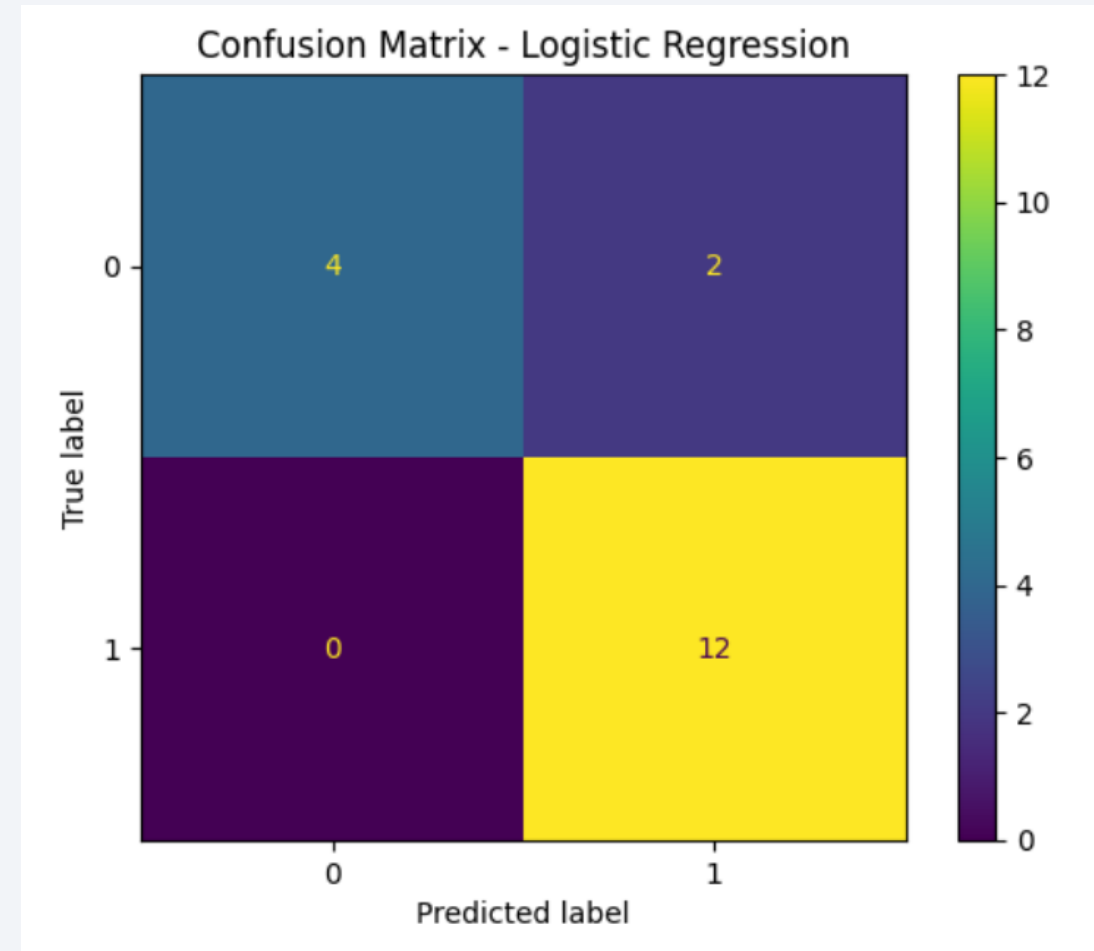
Classification Accuracy

- Logistic Regression and KNN achieve the highest accuracy (0.94)
- SVM and Decision Tree show slightly lower performance (0.89)
- All models perform well, with accuracy close to 90% or higher
- Logistic Regression and KNN are the best choices for this problem



Confusion Matrix

- The model correctly predicts most successful launches (12 true positives)
- No successful launches are misclassified as failures (0 false negatives)
- Some failures are misclassified as successes (2 false positives)
- Overall, the model shows strong performance, especially in detecting successful landings



Conclusions

- Launch success has improved significantly over time, indicating learning and technological advancement
- Certain launch sites (especially KSC LC-39A and CCAFS LC-40) concentrate most successful launches
- Payload mass and orbit type influence outcomes, but success is achieved across a wide range of conditions
- Machine learning models achieve high accuracy (more than 89%), confirming that launch success can be effectively predicted

Appendix

- Python notebooks for data collection, wrangling, EDA, and modeling
- API integration code (SpaceX REST API requests)
- Web scraping scripts using BeautifulSoup
- Data visualization charts (Matplotlib / Plotly / Dash)
- Machine learning models (Logistic Regression, SVM, Decision Tree, KNN)
- Confusion matrices and model evaluation metrics to select best model performance

Thank you!

